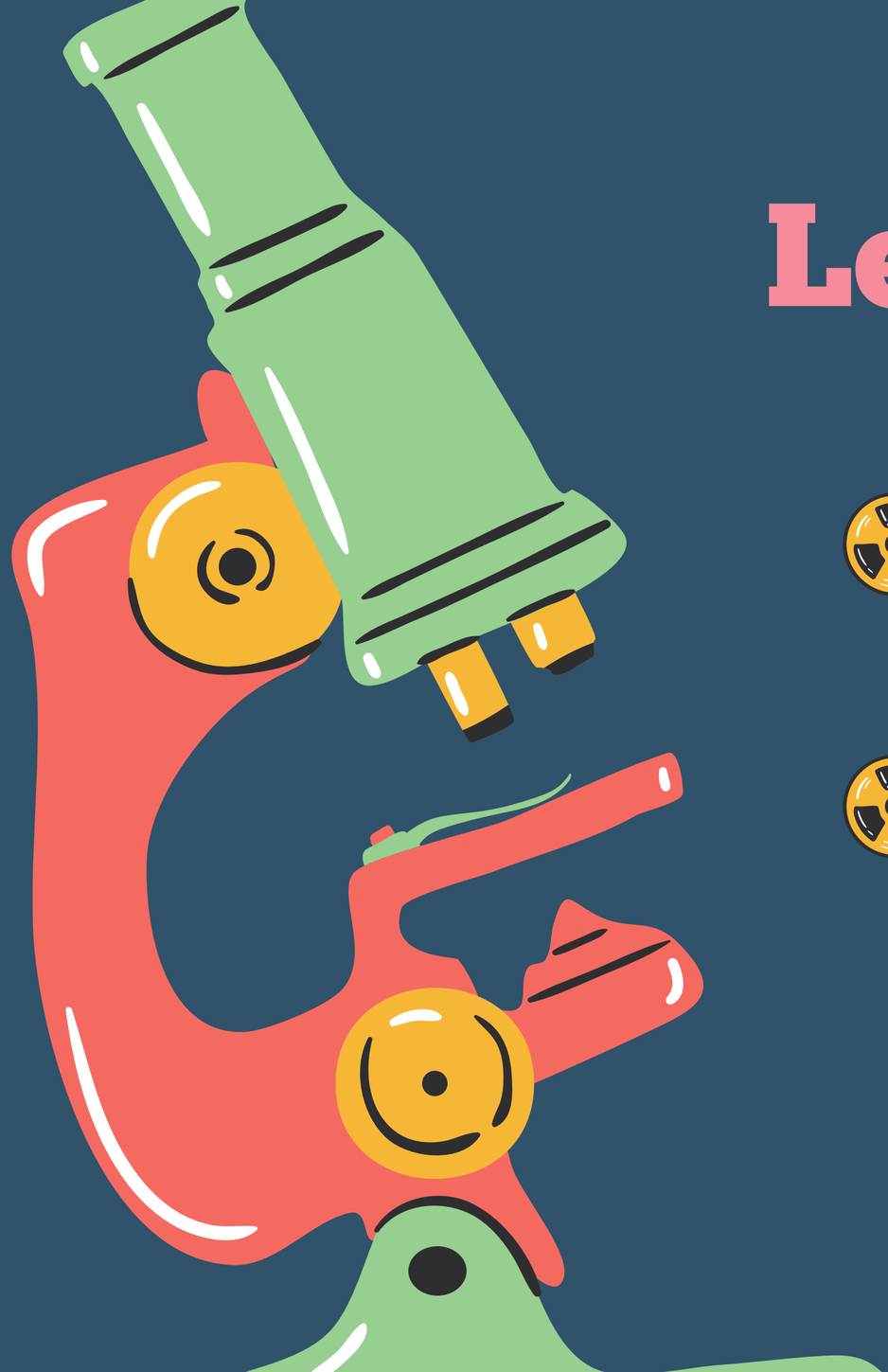


Chemical Names and Formulas

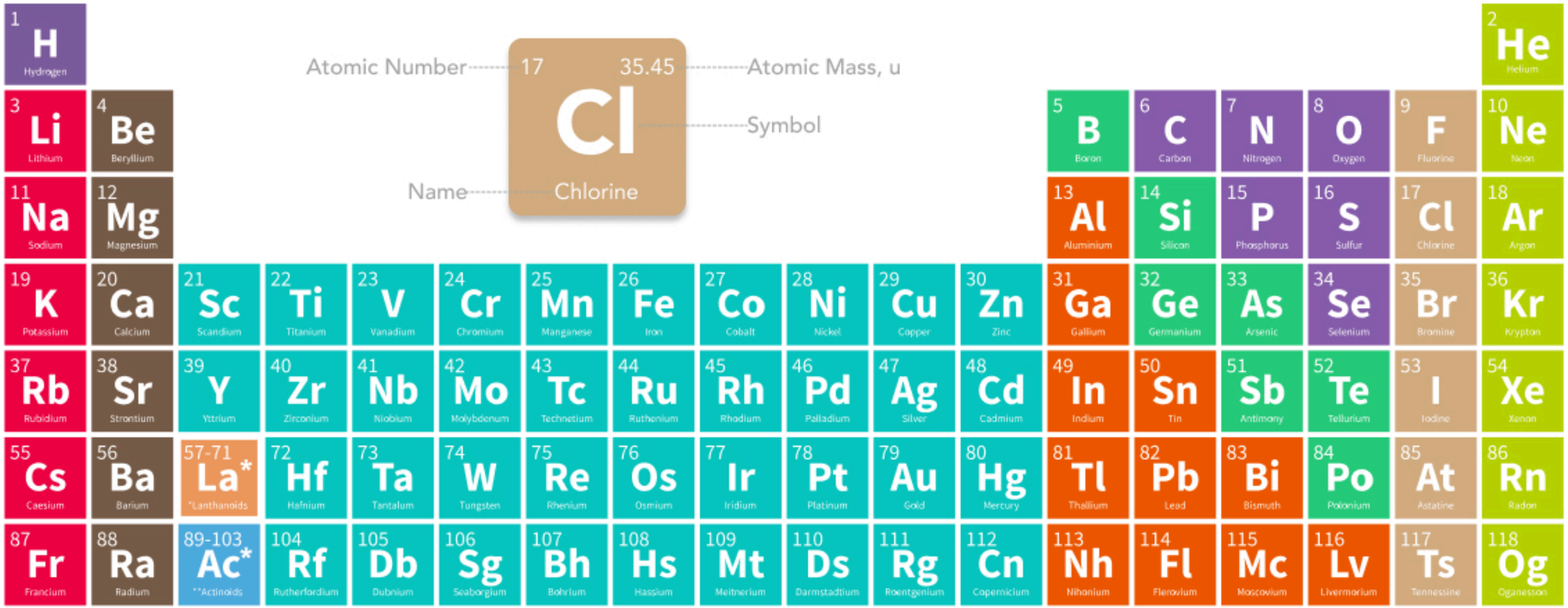


Learning Objectives:

- Recognize the formulas of common chemical substances
- Name some common chemical substances given their formula and write formulas given the names of some common chemical substances.

STANFORD ADVANCED MATERIALS

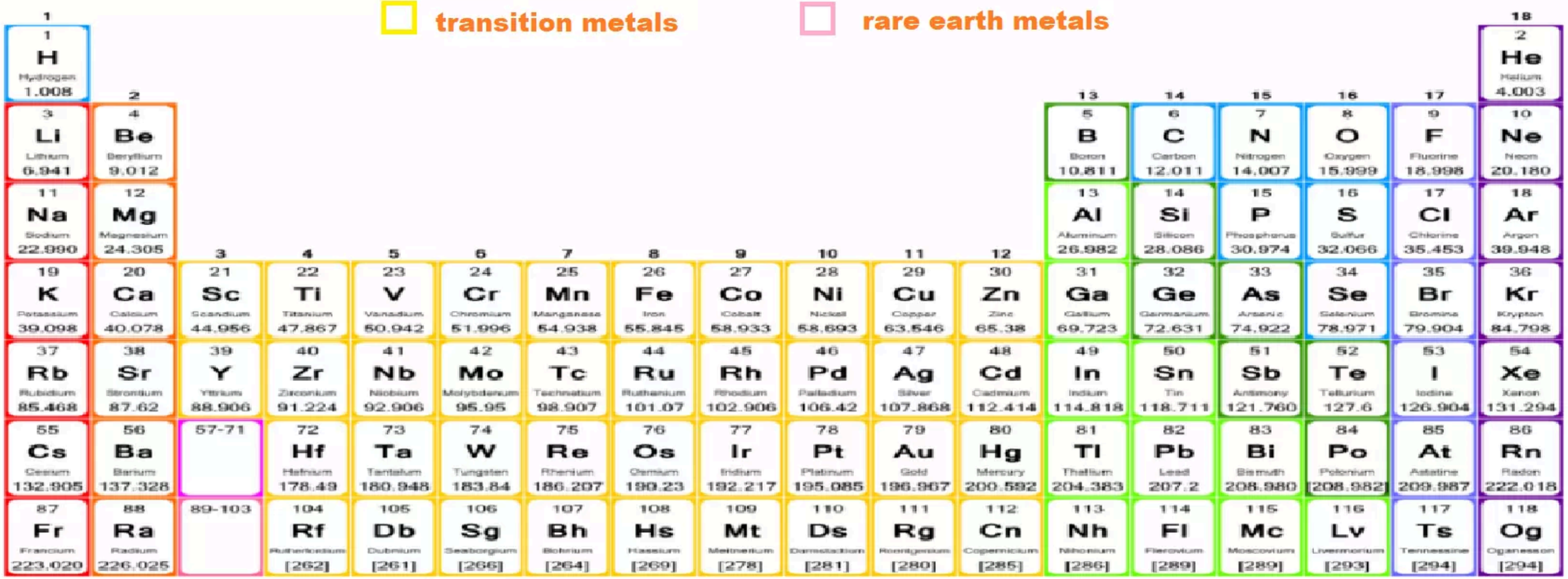
PERIODIC TABLE OF ELEMENTS



*Lanthanoids	57 La Lanthanum	58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
*Actinoids	89 Ac Actinium	90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

The Modern Periodic Table

- hydrogen
- alkali metals
- alkali earth metals
- transition metals
- Poor metals
- nonmetals
- noble gases
- rare earth metals



COMPOUND

Any substance composed of identical molecules consisting of atoms of two or more chemical elements.



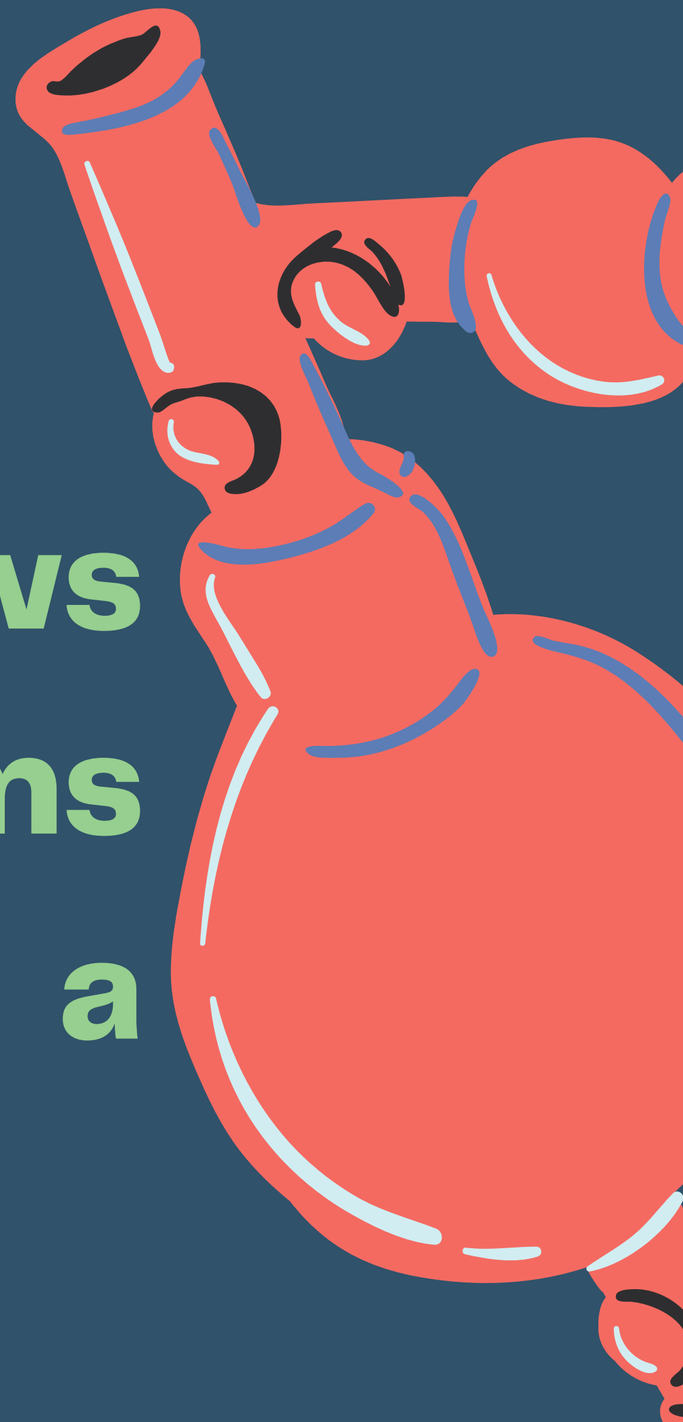
CHEMICAL FORMULA

Identifies each constituent element by its chemical symbol and indicated the proportionate number of atoms of each element.



STRUCTURAL FORMULA

A formula which shows the arrangement of atoms in the molecule of a compound.



LEWIS DOT STRUCTURE

Is a visual representations of the valence electrons in atoms and molecules.



IONIC COMPOUNDS

- Consists of Cations (positive ions) and Anions (negative ions) held together by electrostatic attraction.
- transferring of electrons
- Usually metal and non-metal
- monoatomic ions and polyatomic ions

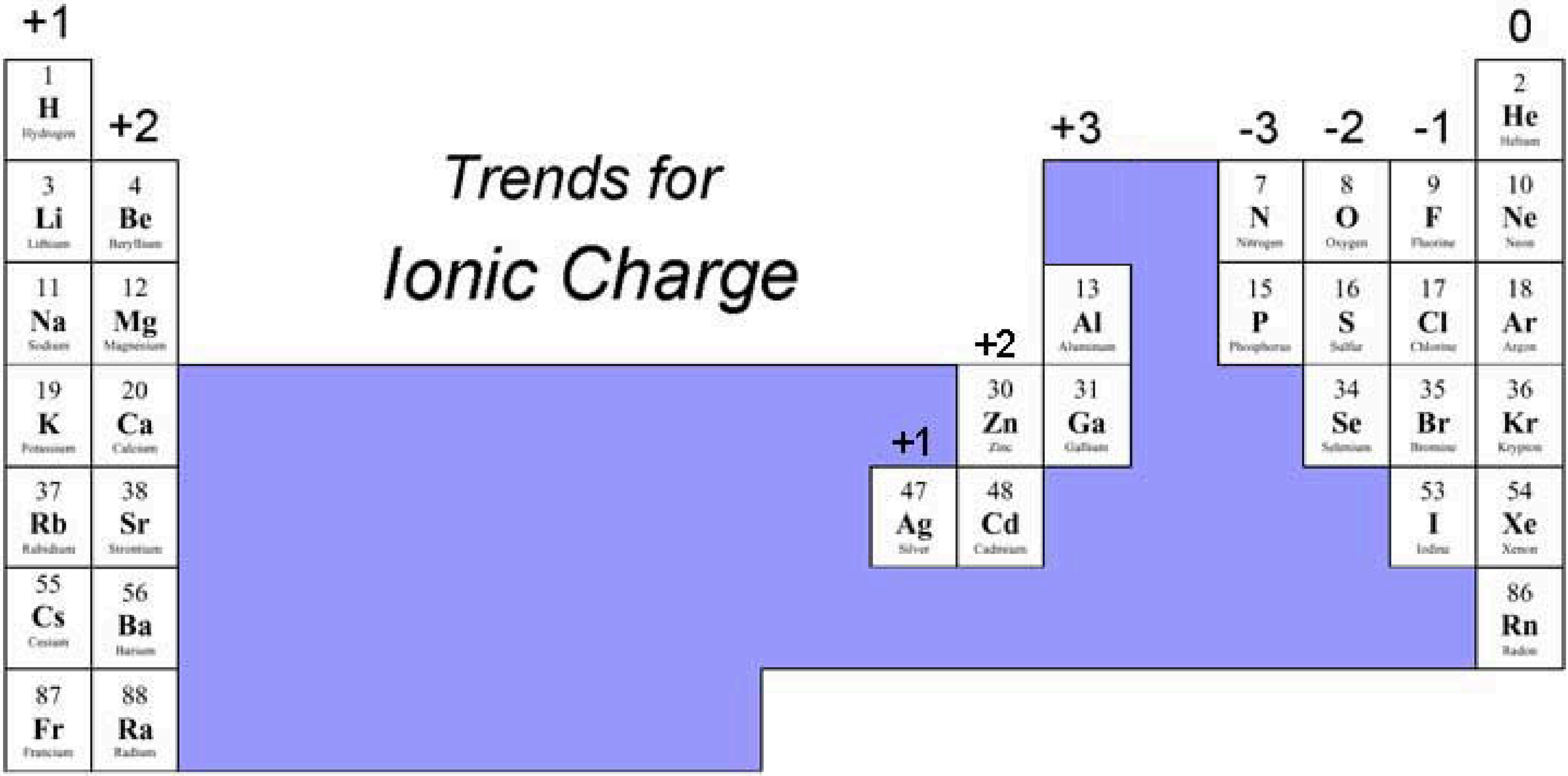


MOLECULAR COMPOUNDS

- Consists of non-metal atoms held together by covalent bonds.
- Sharing of electrons
- Covalent or molecular compounds contain atoms held together by covalent bonds.



Trends for Ionic Charge



MONOATOMIC CATION

"Mono" means one.

"Atomic" refers to atom.

A monoatomic ion is a single atom that has gained or lost electrons, resulting in a charge.

Cation (positive charge) – the atom loses electrons

- Sodium ion: Na^+ (sodium loses 1 electron)
- Calcium ion: Ca^{2+} (calcium loses 2 electrons)

Anion (negative charge) – the atom gains electrons

- Chloride ion: Cl^- (chlorine gains 1 electron)
- Oxide ion: O^{2-} (oxygen gains 2 electrons)

Group IA elements form +1 ions: Li^+ , Na^+ , K^+ , Rb^+ , Cs^+
Group IIA elements form +2 ions: Be^{+2}, Mg^{+2}, Ca^{+2}, Sr^{+2}, Ba^{+2}

silver ion = Ag^{+1}

cadmium ion = Cd^{+2}

aluminum ion = Al^{+3}

zinc ion = Zn^{+2}

HOW WE SHOULD NAME THEM?

element name + ion

MONOATOMIC CATION

1. Sr^{+2}
2. Ga^{+3}
3. Ag^{+1}
4. Ba^{+2}
5. Cd^{+2}

TRANSITION METALS

The Stock system is used to name transition metals, Sn, and Pb that each form more than one ion:

element name (charge in Roman numerals) + ion

Fe^{+2} = _____ Cu^{+} = _____

Fe^{+3} = _____ Cu^{+2} = _____

TRANSITION METALS

1. Cr^{+2}
2. Ti^{+4}
3. Pd^{+2}
4. Sc^{+3}
5. Pt^{+4}
6. Hg^{+2}
7. Sn^{+4}
8. V^{+5}
9. Pb^{+2}
10. Ni^{+3}

NAMING MONOATOMIC ANIONS

When a nonmetal forms an anion

element stem name + "ide" + ion

P = phosphorus atom → P^{-3} = _____

O = oxygen atom → O^{-2} = _____

N = nitrogen atom → N^{-3} = _____

F = fluorine atom → F^{-} = _____

NAMING POLYATOMIC IONS

Ions made up of more than one atom are polyatomic ions

___1. OH^-	A. sulfate ion
___2. SO_4^{-2}	B. acetate ion
___3. NO_3^-	C. nitrite ion
___4. SO_3^{-2}	D. nitrate ion
___5. NO_2^-	E. sulfite ion
	F. hydroxide ion

NAMING POLYATOMIC IONS

Ions made up of more than one atom are polyatomic ions

F	1. OH^-	A. sulfate ion
A	2. SO_4^{-2}	B. acetate ion
D	3. NO_3^-	C. nitrite ion
E	4. SO_3^{-2}	D. nitrate ion
C	5. NO_2^-	E. sulfite ion
		F. hydroxide ion

ite – low number of oxygen

ate – high number of oxygen

Individual ions	Chemical Formula	Chemical Name
Na^{+1} Cl^{-1}		
Al^{+3} S^{-2}		
Fe^{+3} N^{-3}		
Ca^{+2} F^{-1}		

NAMING MOLECULAR FORMULA

Number	Greek Prefix	Number	Greek Prefix
1	mono	6	hexa
2	di	7	hepta
3	tri	8	octa
4	tetra	9	nona
5	penta	10	deca

FOR THE FIRST ELEMENT:

Greek prefix + element name

FOR THE SECOND ELEMENT:

Greek prefix + element name stem + "ide"

Covalent Compounds	Chemical Name
PCl_5	
XeF_6	
N_2O_4	
Cl_2O_7	
P_4O_{10}	

NAMING MOLECULAR FORMULA

1. N_2O_5
2. IC_3
3. SiO_2
4. NCl_3
5. CCl_4
6. Bromine trifluoride
7. Xenon hexafluoride
8. Diiodine pentoxide
9. Tetraphosphorus decoxide
10. Antimony pentachloride